Advanced Bayesian Learning

Mixture models and Bayesian nonparametrics

Lecture 3

Mattias Villani

Department of Statistics
Stockholm University

Topic overview

- Reminder: **Multinomial data** - **Dirichlet prior**
- **Bayesian histograms**
- The **Dirichlet process**
- Beyond DP: **Pitman-Yor** and **Probit stick-breaking**
- **Finite mixture models**
- **Dirichlet process mixtures**
- **MCMC** for Dirichlet process mixtures

The Dirichlet distribution

- $\boldsymbol{\theta} \sim \mathrm{Dirichlet}(\boldsymbol{\alpha})$ with density ($\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$)

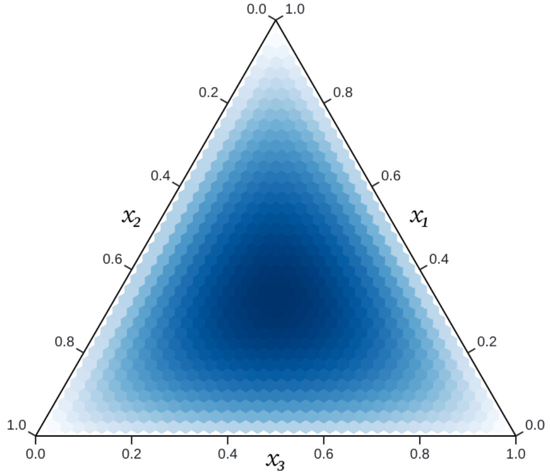
$$p(\theta_1, \theta_2, ..., \theta_k) \propto \prod_{j=1}^{K} \theta_j^{\alpha_j - 1}.$$

- Define $\alpha_0 = \sum_{j=1}^{K} \alpha_j$ and $\bar{\boldsymbol{\alpha}} = \boldsymbol{\alpha}/\alpha_0$.
- Expected value and variance of $\mathrm{Dirichlet}(\alpha_1, ..., \alpha_K)$

$$\mathbb{E}(\theta_j) = \frac{\alpha_j}{\alpha_0} = \bar{\alpha}_j \qquad \mathbb{V}(\theta_j) = \frac{\mathbb{E}(\theta_j)\left[1 - \mathbb{E}(\theta_j)\right]}{1 + \alpha_0}$$

- Note that α_0 is a precision parameter (large α_0, low variance).
- We can also write $\boldsymbol{\theta} \sim \mathrm{Dirichlet}(\alpha_0 \bar{\boldsymbol{\alpha}})$.

Interactive - Dirichlet distribution



Conjugate analysis for multinomial data

- **Data**: $\boldsymbol{n} = (n_1, ..., n_K)$, where $n_k = \#$ items in category k.
- **Prior**

$$\boldsymbol{\theta} \sim \text{Dirichlet}(\boldsymbol{\alpha})$$

- **Likelihood**

$$p(n_1, n_2, ..., n_K | \theta_1, \theta_2, ..., \theta_K) \propto \prod_{k=1}^{K} \theta_k^{n_k}$$

- **Posterior**

$$\boldsymbol{\theta} | \boldsymbol{n} \sim \text{Dirichlet}(n_1 + \alpha_1, ..., n_K + \alpha_K) = \text{Dirichlet}(\boldsymbol{n} + \boldsymbol{\alpha})$$

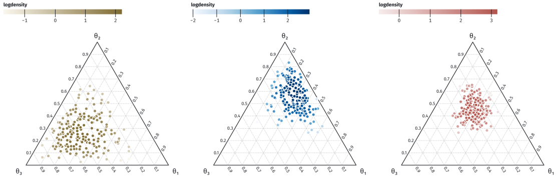
- **Posterior mean**

$$\mathbb{E}(\boldsymbol{\theta} | \boldsymbol{n}) = \frac{\boldsymbol{n} + \boldsymbol{\alpha}}{n + \alpha_0}$$

- Alternatively

$$\boldsymbol{\theta} | \boldsymbol{n} \sim \text{Dirichlet}\left((n + \alpha_0) \frac{\boldsymbol{n} + \boldsymbol{\alpha}}{n + \alpha_0} \right)$$

Interactive - Multinomial model



Bayesian histograms

- **Partition** the data space into disjoint intervals (**bins**): $B_1, B_2 \ldots, B_K$. Width of B_k is Δ_k.
- The usual histogram estimate

$$\hat{p}(x) = \frac{n_k / n}{\Delta_k}$$

- **Probability model** for **histograms**

$$p(x) = \sum_{k=1}^{K} \frac{\omega_k}{\Delta_k} 1(x \in B_k)$$

- n_k = number of obs in B_k. Multinomial data.
- **Prior** on $\boldsymbol{\omega} = (\omega_1, ..., \omega_K)$

$$\boldsymbol{\omega} \sim \text{Dirichlet}(\boldsymbol{\alpha}) = \text{Dirichlet}(\alpha_0 \bar{\boldsymbol{\alpha}})$$

- **Posterior**

$$\boldsymbol{\omega} | \boldsymbol{n} \sim \text{Dirichlet}\left((n + \alpha_0) \frac{\boldsymbol{n} + \boldsymbol{\alpha}}{n + \alpha_0} \right)$$

- total counts = data counts + prior counts.

Bayesian histograms

- **Prior**

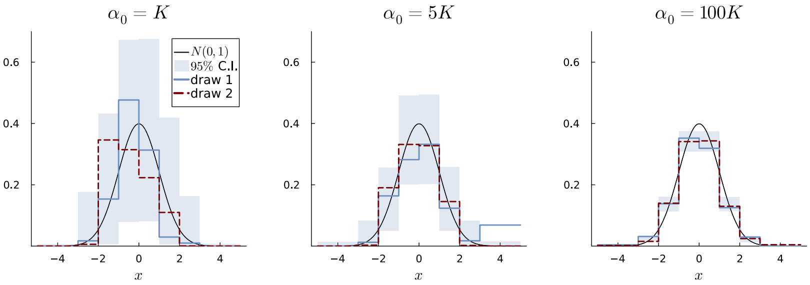
$$\mathrm{Dirichlet}(\alpha_0 \bar{\boldsymbol{\alpha}})$$

- Specify prior mean $\bar{\boldsymbol{\alpha}}$ from a **base distribution** P_0:

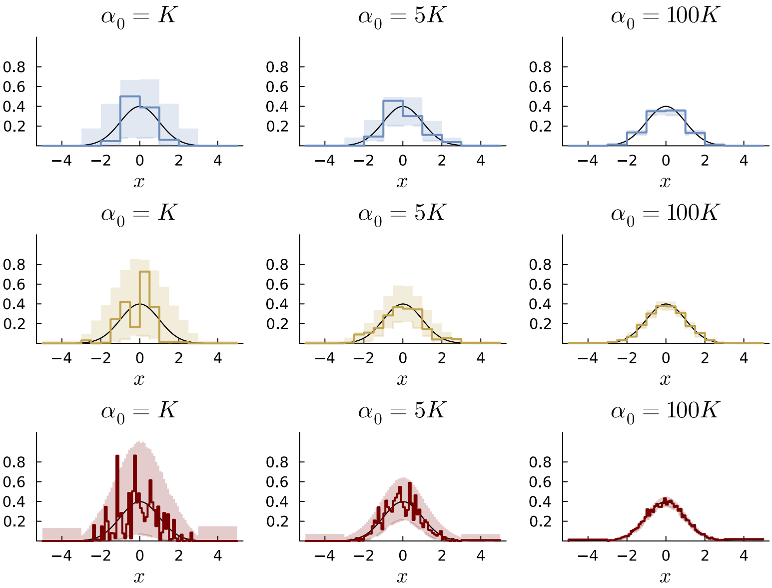
$$\bar{\boldsymbol{\alpha}}_k = P_0(B_k) = \Pr(x \in B_k).$$

- Specify the strength of prior beliefs by α_0.
- Properties of Dirichlet prior:
  - **easy computations**
  - **easy to specify hyperparameter** $\bar{\boldsymbol{\alpha}}$ and α_0.
  - **no smoothness**: adjacent bin are negatively correlated.
  - **sensitive** to the choice of **bins**.

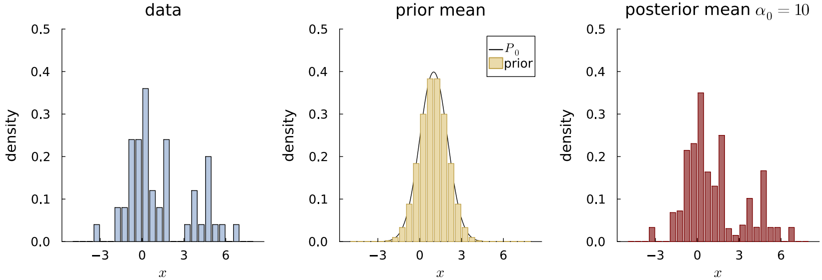
Simulation from histogram prior



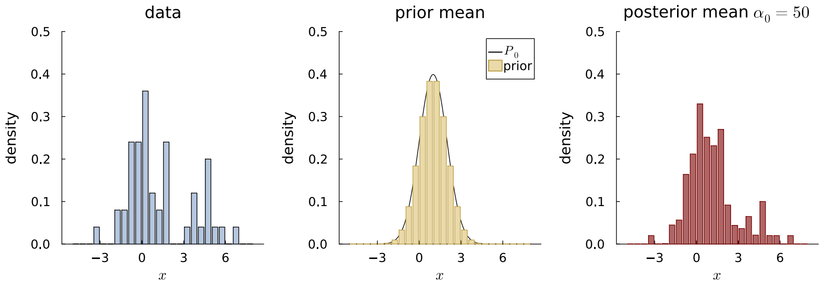
Simulation from histogram prior - bin size



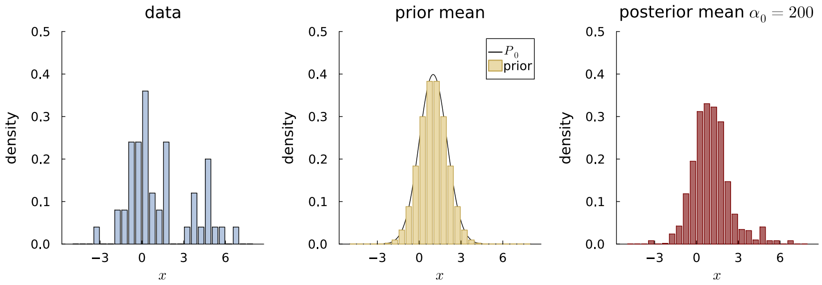
Example with $n = 50$ and $\alpha_0 = 10$



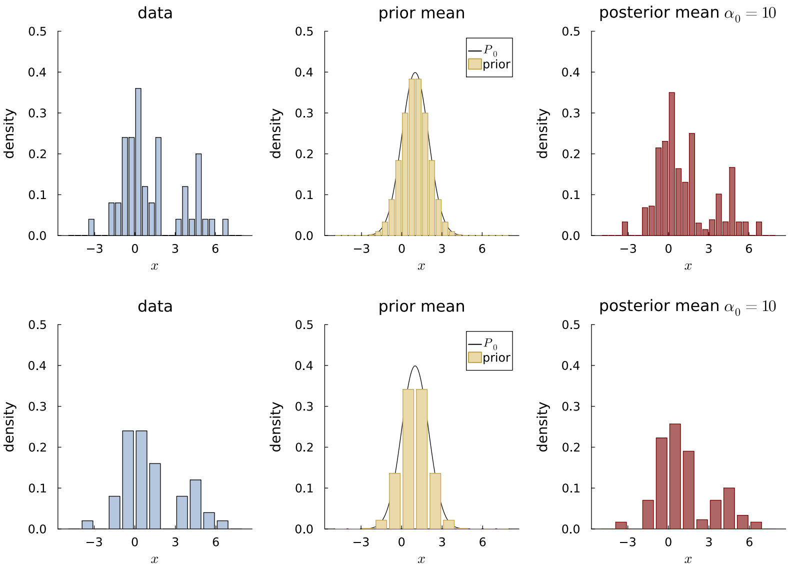
Example with $n = 50$ and $\alpha_0 = 50$



Example with $n = 50$ and $\alpha_0 = 200$



Histograms are sensitive to the choice of bins



The Dirichlet process

- Let $B_1, B_2, ..., B_k$ be a partition of the outcome space Ω.
- $P(B_1), ..., P(B_k)$ denotes the distribution over the partition.
- Dirichlet distribution is a **distribution over distributions**:

$$(P(B_1), ..., P(B_k)) \sim \mathrm{Dirichlet}(\alpha_0 P_0(B_1), ..., \alpha_0 P_0(B_k))$$

  where P_0 is a fixed probability measure (e.g. $N(0, 1)$).

- Dirichlet is **closed under summation or splitting** of bins.
  $\Rightarrow$ consistent definition of a **stochastic process**. c.f. GPs.

- A random probability measure P follows a **Dirichlet process** $P \sim \mathrm{DP}(\alpha_0 \cdot P_0)$ with **base measure** P_0 iff

$$(P(B_1), ..., P(B_k)) \sim \mathrm{Dirichlet}(\alpha_0 P_0(B_1), ..., \alpha_0 P_0(B_k))$$

  for any finite measureable partition $B_1, ..., B_k$.

The Dirichlet process - properties

- If $P \sim \mathrm{DP}(\alpha_0 P_0)$ then

$$P(B) \sim \mathrm{Beta}\left[\alpha_0 P_0(B), \alpha_0\left(1 - P_0(B)\right)\right], \text{ for any } B \in \mathcal{B}$$

$$\mathbb{E}\left[P(B)\right] = P_0(B)$$

$$\mathbb{V}\left[P(B)\right] = P_0(B)\left[1 - P_0(B)\right]/(1 + \alpha_0)$$

- **Model**

$$y_i | P \overset{\mathrm{iid}}{\sim} P \text{ , for } i = 1, ..., n$$

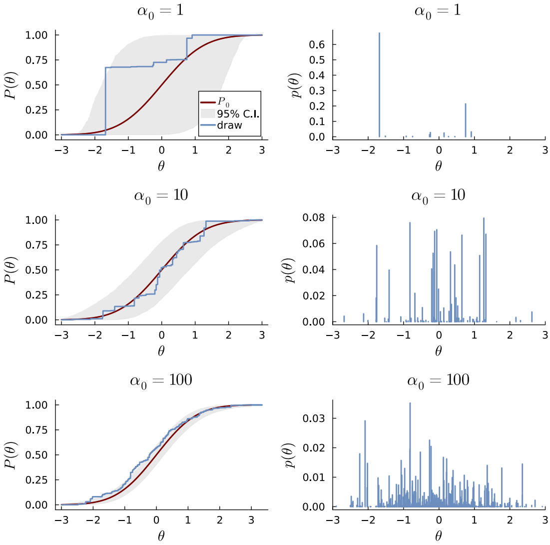
- **Prior**

$$P \sim \mathrm{DP}(\alpha_0 P_0)$$

- **Posterior** for a **finite partition**, $P(B_1), ..., P(B_k)|\mathrm{y}$ is

$$\mathrm{Dirichlet}\left(\alpha_0 P_0(B_1) + \sum_{i=1}^{n} 1_{y_i \in B_1}, ..., \alpha_0 P_0(B_k) + \sum_{i=1}^{n} 1_{y_i \in B_k}\right)$$

Dirichlet process realizations



The Dirichlet process - properties

- **Posterior** for the unknown probability distribution P

$$P|y_1, \ldots, y_n \sim \mathrm{DP}\left(\alpha_0 P_0 + \sum_{i=1}^{n} \delta_{y_i}\right)$$

- Since

$$P(B) \sim \mathrm{Beta}\left(\alpha_0 P_0(B) + \sum_{i=1}^{n} 1_{y_i \in B}, \alpha_0(1 - P_0(B)) + \sum_{i=1}^{n} 1_{y_i \in B^c}\right)$$

so

$$\mathbb{E}\left(P(B)|y_1, \ldots, y_n\right) = \left(\frac{\alpha_0}{\alpha_0 + n}\right) P_0(B) + \left(\frac{n}{\alpha_0 + n}\right) \sum_{i=1}^{n} \frac{1}{n} \delta_{y_i}(B)$$

Estimating a distribution function with a DP prior

- If $B = (-\infty, y]$ then

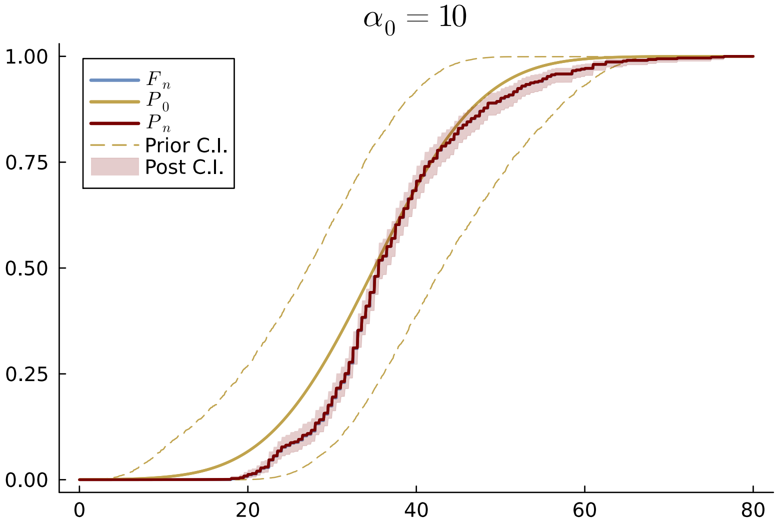
$$E\left(F(y) | y_1, \ldots, y_n\right) = \left(\frac{\alpha_0}{\alpha_0 + n}\right) F_0(y) + \left(\frac{n}{\alpha_0 + n}\right) F_n(y)$$

  - $F(y)$ is the unknown cdf
  - $F_0(y)$ is the cdf from P_0
  - $F_n(y) = \frac{1}{n} \sum 1_{y_i \leq y}$ is the empirical cdf

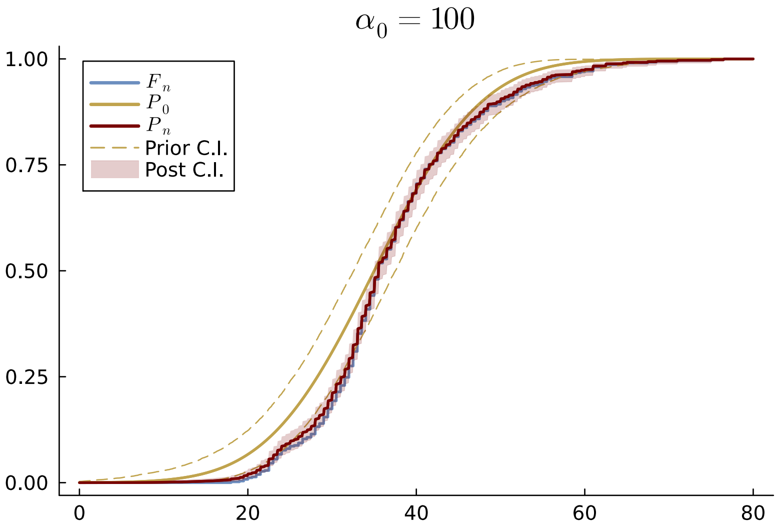
- $F(\cdot)$ is **discrete with probability one** in the DP posterior.
- **Realisations from a DP are discrete with probability one**.

  - Clearly a bad property for continuous data ...
  - But very useful for clustering (mixture models).

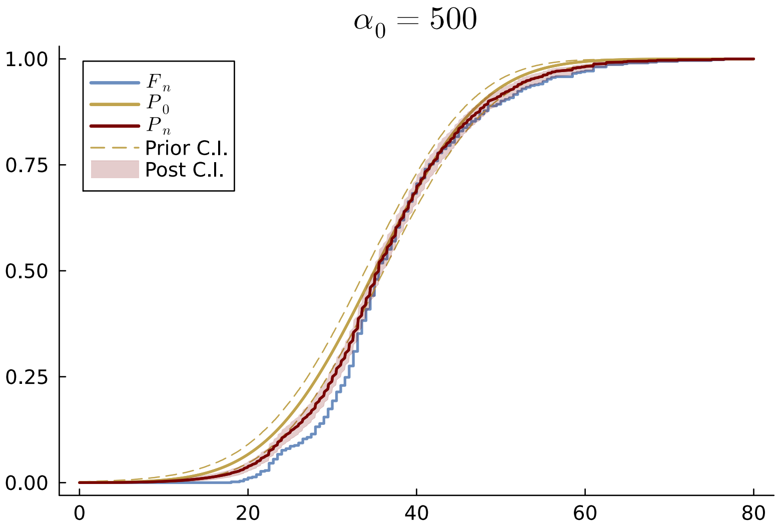
Fish length data $\alpha_0 = 10$



Fish length data $\alpha_0 = 100$



Fish length data $\alpha_0 = 500$



Stick-breaking characterization of the DP

- $P \sim DP(\alpha_0 P_0) \equiv$ **infinite mixture of point masses**

$$P(\cdot) = \sum_{h=1}^{\infty} \pi_h \delta_{\theta_h}$$

$$\pi_h = V_h \prod_{\ell<h} (1 - V_\ell)$$

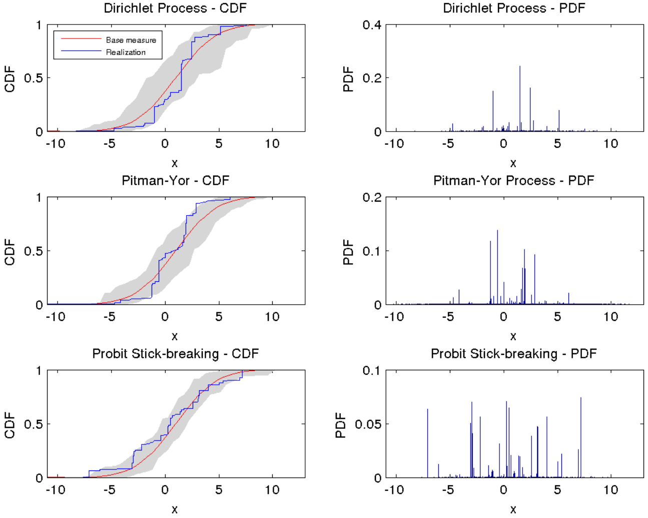
$$V_h \overset{\text{iid}}{\sim} \text{Beta}(1, \alpha_0)$$

$$\theta_h \overset{\text{iid}}{\sim} P_0$$

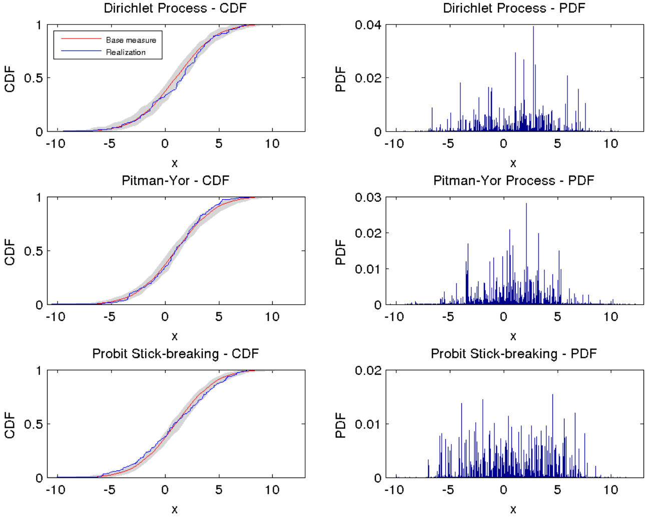
- Alternative notation for $P \sim DP(\alpha_0 P_0)$:

$$\pi = (\pi_1, \pi_2, ...) \sim \text{Stick}(\alpha_0) \text{ and } \theta_h \sim P_0$$

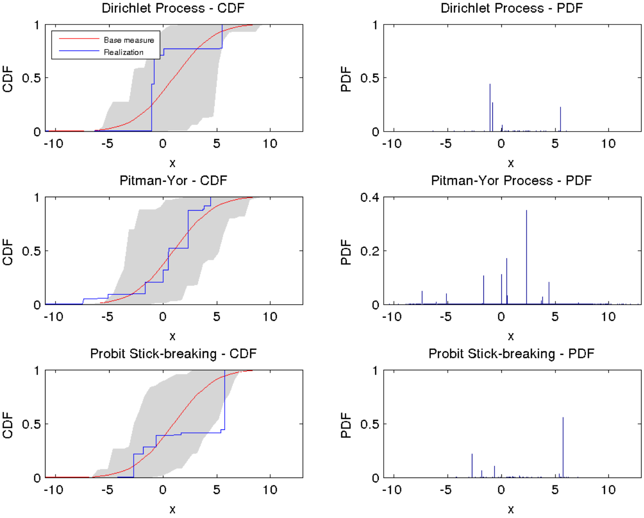
Simulating stick-breaking $\alpha_0 = 10, P_0 = N(1, 3^2)$



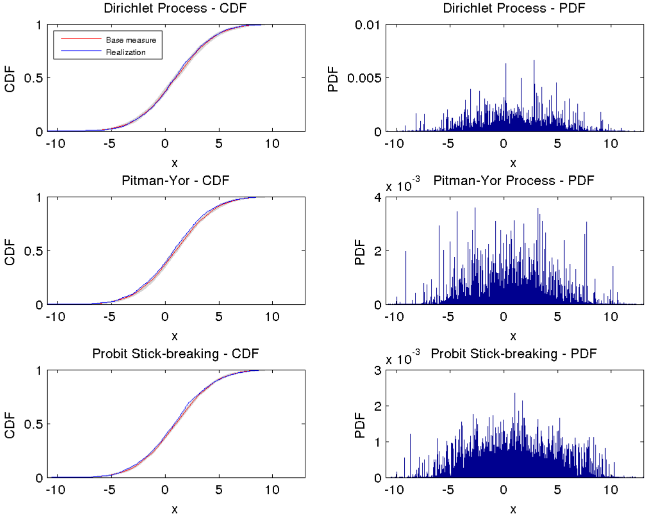
Simulating stick-breaking $\alpha_0 = 100$, $P_0 = N(1, 3^2)$



Simulating stick-breaking $\alpha_0 = 1, P_0 = N(1, 3^2)$



Simulating stick-breaking $\alpha_0 = 1000,\ P_0 = N(1, 3^2)$



Beyond DP - Pitman-Yor and Probit sticks

- **Pitman-Yor process** with parameters P_0, $0 \leq a < 1$ and $b > -a$:

$$P(\cdot) = \sum_{h=1}^{\infty} \pi_h \delta_{\theta_h} \quad \theta_h \stackrel{iid}{\sim} P_0$$

$$\pi_h = V_h \prod_{\ell<h} (1 - V_\ell)$$

$$V_h \stackrel{iid}{\sim} Beta(1-a, b+ha)$$

- **Probit stick-breaking** with parameters μ and σ:

$$P(\cdot) = \sum_{h=1}^{\infty} \pi_h \delta_{\theta_h} \quad \theta_h \stackrel{iid}{\sim} P_0$$

$$\pi_h = V_h \prod_{\ell<h} (1 - V_\ell)$$

$$V_h = \Phi(x_h), \text{ where } x_h \stackrel{iid}{\sim} N(\mu, \sigma^2)$$